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REVISED STANDARD OISD-STD-226 (JUNE 2024)

CROSS COUNTRY NATURAL GAS PIPELINES AND CITY GAS DISTRIBUTION NETWORKS

Inception	August 2007
Amendment	October 2013
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OISD-STD-226 (2024) is comprised of Two Parts

PART-I

- FOR CROSS COUNTRY PIPELINE
(DESIGN, OPERATION & MAINTENANCE)

PART-II

- FOR CITY GAS DISTRIBUTION (CGD) NETWORK

Flow of Presentation

Changes in OISD-STD-226 Part-I

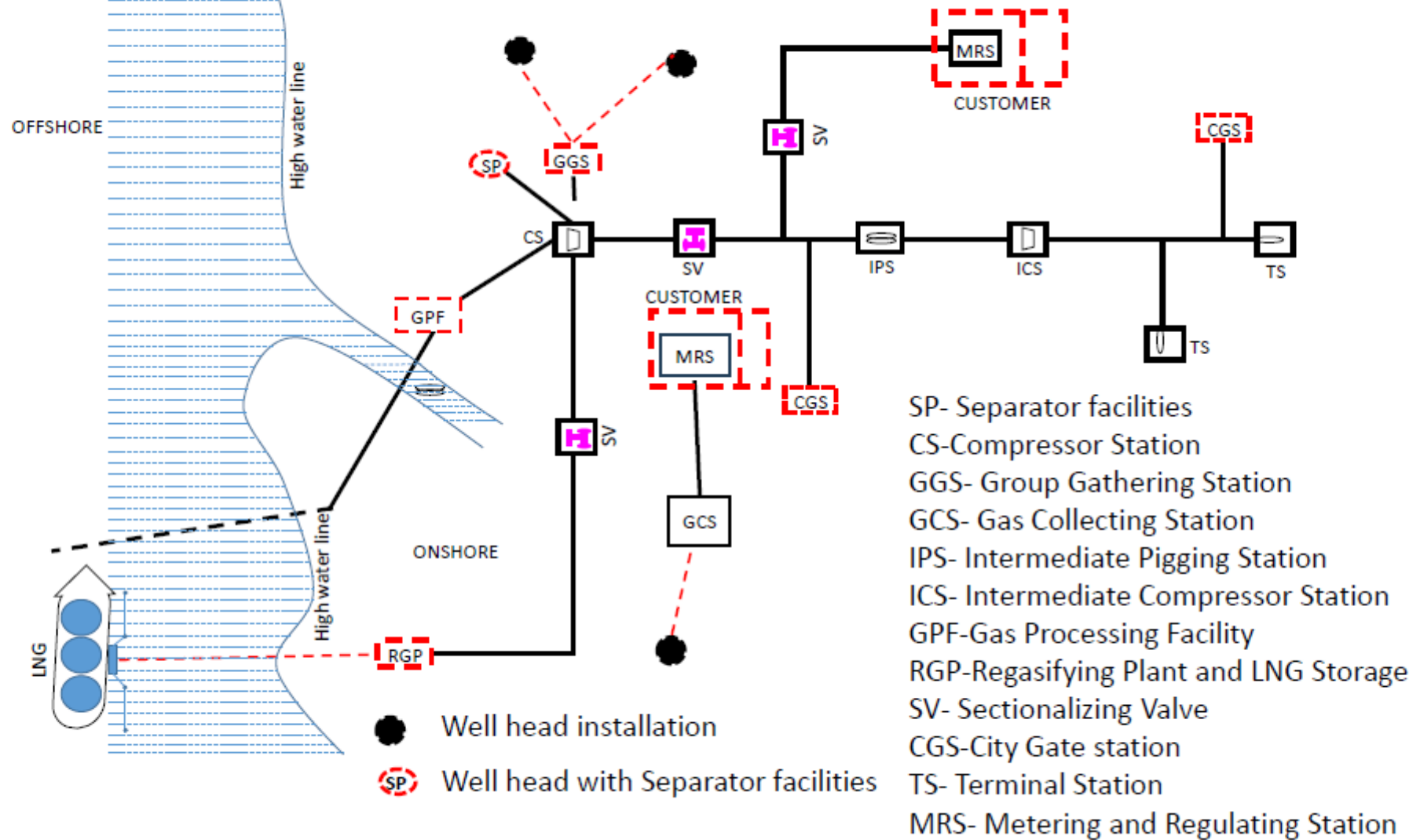
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- SCOPE
 - RISK ASSESSMENT & LOCATION CLASS
 - PIPELINE DESIGN
 - SECTIONALIZING VALVE STATION
 - PIGGING FACILITY & FREQUENCY
 - SCADA, PIDS, COMMUNICATION, CCTV
 - CATHODIC PROTECTION FOR PIPELINE
 - PATROLLING OF PIPELINE
 - ELECTRICAL MITIGATION

Scope of OISD-STD-226 Part-I

This standard outlines the minimum requirement for safety in design, engineering, materials, construction, testing, commissioning, operation, maintenance, corrosion protection, integrity management, modifications, abandonment, safety and fire protection systems of onshore cross-country natural gas pipelines including offshore section of a predominately onshore natural gas pipeline (such as creek / estuary crossings) and spur lines including associated facilities and installations as outlined in **Annexure-1** and as mentioned below.

- Feeder pipeline from Group Gathering station (GGS)/ Gas collecting station (GCS) or equivalent after separation.
- Transmission/ sub transmission pipeline.
- Spur/ Branch pipeline.
- Dedicated pipeline to consumer after separation facility.

Annexure-1 (Clause 2.0) Scope of this Standard



Facilities indicated by Solid Lines are under the scope of this Standard

Risk Assessment

Cl.
5-3

The risk management process should make use of the GIS database of the pipeline and other pertinent data such as population data, environmental data etc.

Cl.
5-3

The risk management process during the project planning and design stage shall enable the selection of most appropriate route of the pipeline and location & layout of facilities besides minimizing and mitigating material and construction risks.

Cl.
5-3

The risk management process during service life should primarily focus on the threats to pipeline integrity and reducing the probability of failure.

Pipeline Design

Pipeline Hydraulic Analysis:

Cl.
5.4

Hydraulic analysis should be analysed to demonstrate that the system can safely transport the fluids for the defined extent of the pipeline system, meeting the functional requirements and applicable regulations.

Corrosion Allowance:

Cl.
5.8.2

Minimum corrosion allowance of **1 mm** has been considered irrespective of internal corrosion mitigation measures considering acceptable internal corrosion rate of 1 mpy.

Cl.
5.8.3

D/t ratio more than 100 shall not be allowed in the cross-country pipelines.

Additional Requirement

Cl.
5.10

Gas quality shall meet requirement as specified in OISD-STD-188.
Monitoring of Moisture, H₂S and CO₂ and O₂ content to be ensured.

Cl.
5.10

In case of pipelines requiring to transport gas for short distances (Up to 10 Kms length) and **not meeting the specification** given in OISD-STD-188, a **detailed and documented corrosion management plan** shall be designed and implemented.

Cl.
5.10

Online moisture, H₂S and CO₂ Analyzers shall be installed at Gas Dispatch terminals receiving gas from a supplier / process plant and quality of gas monitored on round-the-clock basis.

Plastic Pipeline

Cl.
6.6

Reinforced Thermoplastic pipes (RTP) or Multi Layered Thermoplastic Pipes, tubes and fittings may be used in pipeline transportation of natural gas in case of isolated / marginal fields / customers, and not forming part of a high-pressure network subject to the conditions as per clause 6.6

Sectionalizing & Station Valve

Cl.
5.9.9

Sectionalizing Valve: Wherever remote operation is not feasible, operation of valve by trained personnel shall be ensured.

Location Class 4, shall have remote operation capability from control room of the network.

Cl.
5.12.1

Station Isolation valve:

Pressure equalization line with valves should be provided for valves 8" and above & pressure above 7 bar.

Cl.
6.3

Carbon Steel Valves: All valves forming part of the pipeline system shall conform to API 6D or equivalent standard. The selection of seat type, material and / or coating /plating should be made based on the type of service viz., clean vs dirty or abrasive, corrosive service etc.

All valves shall be of 'Fire Safe' design in accordance with API 607 or API 6FA or equivalent

Layout

Cl.
5.11.2

Hazard identification and Risk Assessment tools and techniques shall be employed to establish the layout scheme along with drawings.

Cl.
5.11.2

Buried sections of station piping **should be avoided.**

Piping required to cross internal roads or others should be provided with **culverts or overhead pipe trestles** with access for inspection and maintenance.

Pigging Facilities

Cl.
5.10

All pipelines should be designed and constructed to be piggable such that **In-Line Inspection (ILI)** tool can pass through the pipeline.

Cl.
5.12.2

Vents, drains and pressure gauges shall be provided on **both minor and major** barrels to check thorough evacuation of trapped fluids unhindered by possible plugging by pipeline debris.

Pigging Facilities

Cl.
5.12.2

Pig signallers (minimum one pig signaller on the minor barrel and one on the pipeline side, preferably non-intrusive type), pressure gauges and pressure safety valve to safeguard against raise in pressure of trapped fluids to unsafe levels etc.

Cl.
5.12.2

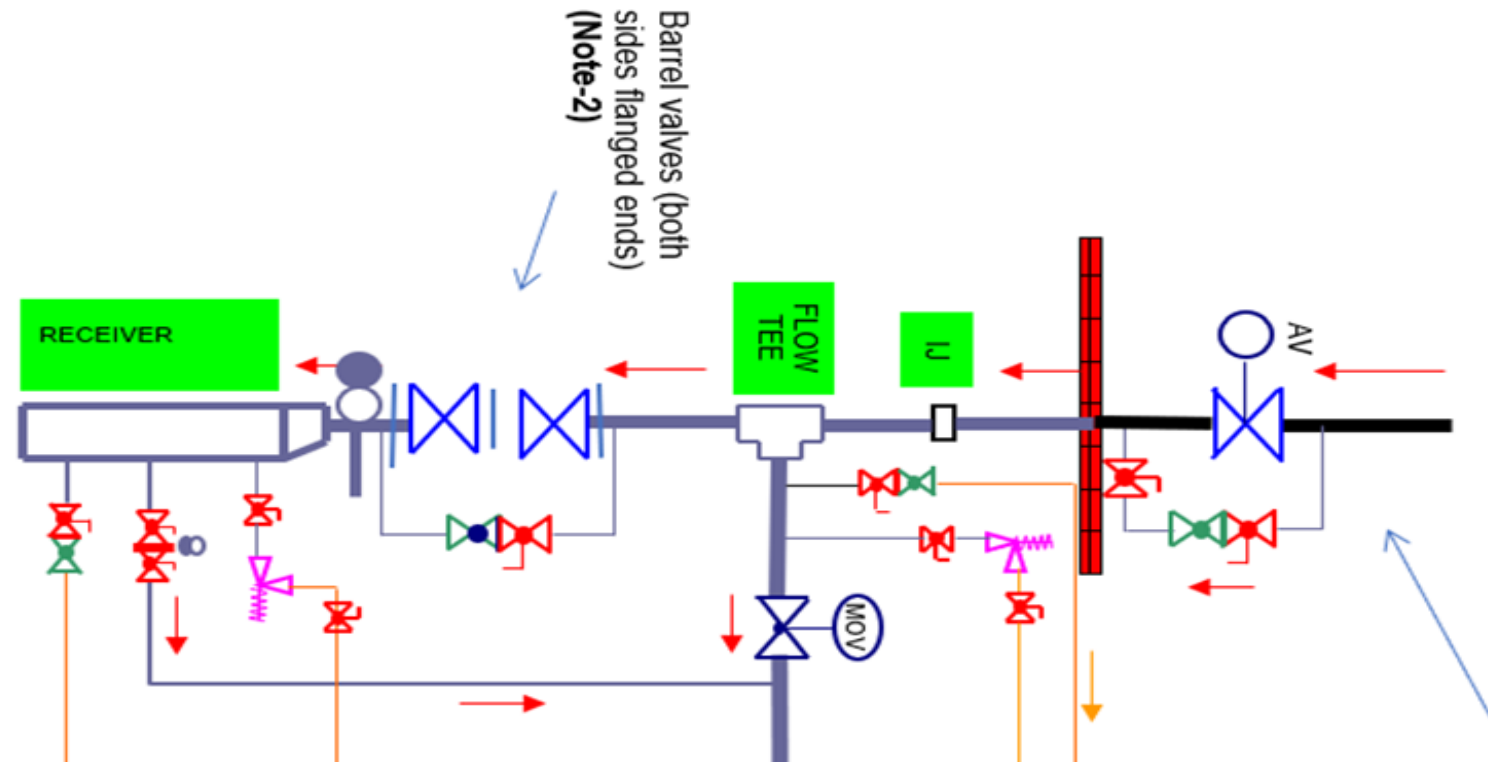
The Pigging facilities should be preferably constructed for **bi-directional pigging**

Cl.
5.12.2

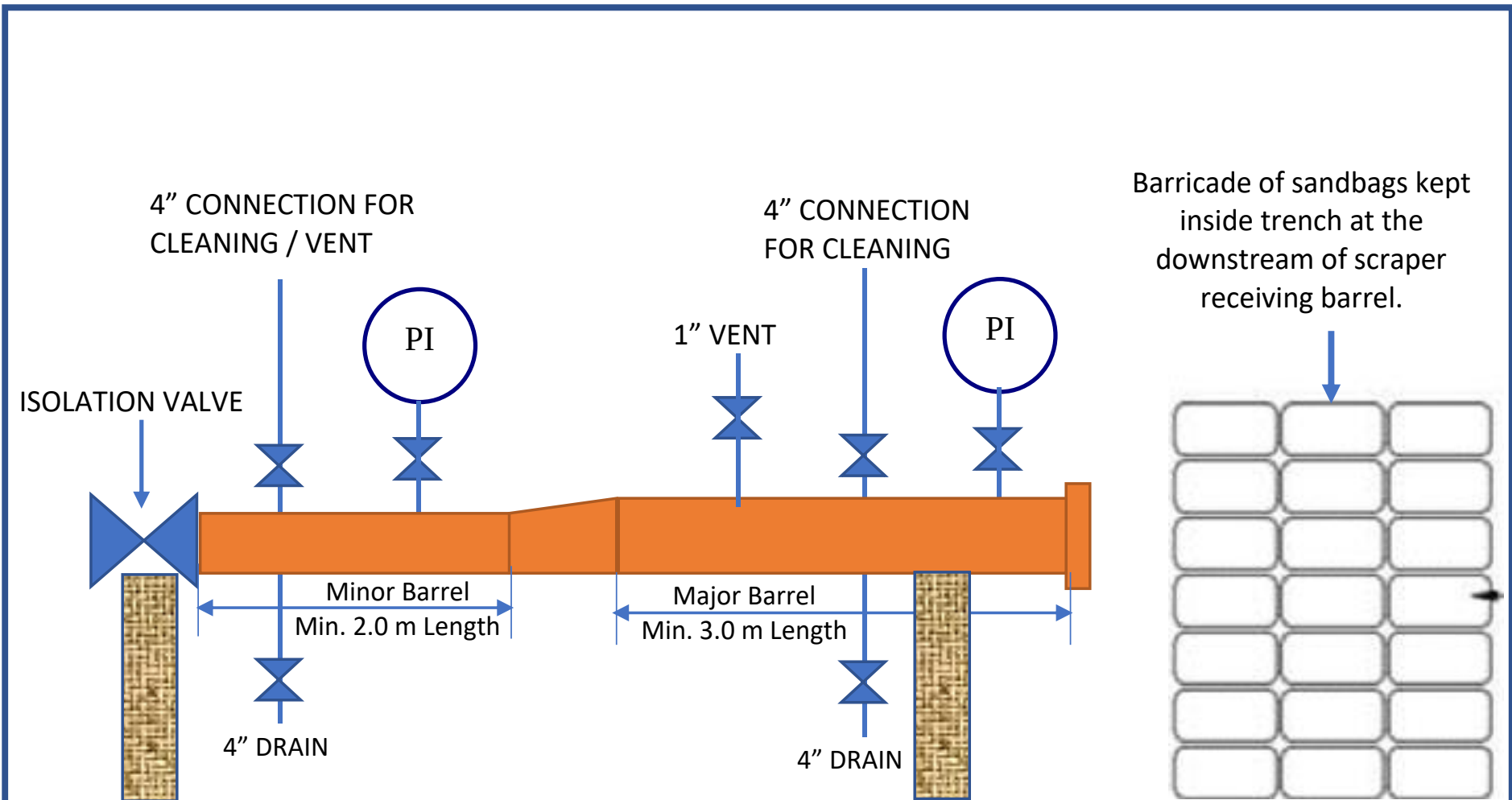
The scrapper barrel **should be designed for continuously pressurized** state with due consideration of pressure relief due to temperature change.

CI.
5.12.2

In each pigging station, a **second mainline valve** of same or similar type with flange ends should be considered for pig receiving / launching barrel with **an intermediate break-flange between the second valve and barrel** with a spacer / blind for isolating the pig barrel for maintenance. (As shown below)



Annexure 4: Schematic for Cleaning Pig Receiver Barrel during Construction



Note: This is a typical schematic for reference to facilitate safe pig retrieval. The operator may modify as per the actual conditions.

Pigging & ILI Frequency

Cl.
11.6

Cleaning Pigging:

Pipeline pigging to be carried out more frequently after commissioning of new pipeline and at interval of **once in 3 years but not more than once in 5 years** in case of pipelines carrying gas of **quality** as specified in clause No. 5.1 or **else once in a year** in case of pipelines carrying gas not meeting as per this clause.

Cl.
11.7

ILI : Gas quality meeting 5.1 : Once in **10 years**.

Gas quality not meeting 5.1 : Once in **5 years**.

Pipeline of age over 25 Years: Once in **8 years**.

If ILI is **not feasible** due to low pressure/ flow, alternate methods of integrity assessment such **as DA, Hydrotest etc.** shall be carried out at required intervals..

Scada & Communication

Cl.
5.17

A reliable and dedicated communications system preferably with **a redundant path for reliability** shall be provided for voice and data communication between various pipeline stations, local / remote / central control rooms and data centres.

Data back-up for minimum **60 days** shall be maintained.

Cl.
5.17

Consideration shall be given to **cybersecurity threats for the SCADA** network. The software shall comply to cyber security laws, government notifications, guidelines, and international standards such as IEC 62443.

Cl.
5.17

(SCADA) with **active and hot standby Control** system or in case of minor pipeline networks suitable alternative arrangement **shall be provided.**

Reliable power systems with battery back-up shall be provided

ESD & CCTV

Cl.
5.14.2

ESD: Emergency shutdown system **(Hardwired)** shall be operable at least from two locations - safe distance - which one should be in the field outside the compressor building and **another in Control Room of compressor station.**

Cl.
5.11.3

CCTV CCTV surveillance system **should be provided for compressor stations, sectionalizing valves, intermediate pigging stations,** gas terminal stations for monitoring movement of people and process facilities. **Back up of minimum 30 days shall** be kept for the CCTV footage.

Patrolling & Crossing Inspection

Cl.
11.5

Pipeline Patrolling:

Emerging techniques like satellite imagery, other aerial surveillance (such as drones) techniques, Pipeline Intrusion detection system(PIDS) etc., should be used to supplement the pipeline corridor surveillance based on threat perception.

Cl.
11.4

Inspection of water body crossings:

Monitoring of water body crossings which are prone to changes in profile/ course should use latest techniques such as Google images/ Drone survey/ SONAR etc. for deciding on proactive measures w.r.t integrity of pipeline due to exposure and washout.

Interference Study, Inspection, Preservation of Pipeline

Cl.
7.4

AC & DC interference study using computer modelling and simulation should be carried out **during design stage of pipeline**, considering steady state and fault conditions and further implementation of mitigation shall be carried out along with the pipeline laying.

Cl.
8.2 & 9.2

Inspection provisions have been specified to ensure good quality and workmanship during pipe manufacturing, construction and testing stages.

Pressure Testing : Test pressure has been increased to **1.5 times** of Design pressure in **class 3 & 4 locations**.

Cl.
9.4 & 9.5

Preservation of pipeline: Preservation pressure after hydrotesting and drying has been defined at **0.5 barg**.

Drying level has been specified based on moisture dew point inside the pipeline system at **minimum -20° C**.

Documentation & PIMS

Cl.
9.6

Documentation:

GIS database shall be used in various drawings like route maps, alignment sheet, crossing drawings, station layouts etc.

Cl.
12.1

PIMS :

All the operating companies shall implement and comply with a documented IMS as above for the pipeline system in accordance with Petroleum and Natural Gas Regulatory Board (Integrity Management System for Petroleum Products Pipelines) Regulations, 2021

OISD-STD-226 (2024) PART-II

PART-II

FOR CITY GAS DISTRIBUTION NETWORKS

FLOW OF PRESENTATION

CHANGES IN OISD-STD-226 Part-II

- 
- SCOPE
 - LAYOUT, PIPING & VALVES
 - CONSUMER INSTALLATIONS AND PROXIMITY TO PROPERTIES
 - PNG ODORIZATION
 - FIRE PROTECTION FACILITIES
 - DEPTH OF COVER & GIS IMPLEMENTATION
 - LEAKAGE SURVEY AND INTEGRITY MANAGEMENT

SCOPE OF OISD-STD-226 PART-II

This standard lays down the minimum safety requirements for design, materials, installation, inspection and testing, commissioning, operation, maintenance & modifications of PNG Distribution in cities/towns/villages for all existing and new pipelines. It covers the equipment, pipelines, accessories etc. installed between Up- Stream Inlet Isolation Valve of City Gate Station (CGS) and;

- a) Up to downstream valve of MRS/custody meter.
- b) Rubber hose after First isolation valve down stream of Consumer Meter for domestic consumer.
- c) Piping and facilities up to the burner of gas Oven and /or any Gas appliances for domestic consumer.
- d) The facilities up-stream of CGS Inlet Valve are covered under Part-1 of this standard.
- e) CNG Dispensing facilities are covered in OISD-STD-179.

Note: Sub transmission pipeline is covered in Part-1 of this document.

Layout, Piping & valves

Cl.
5.0

Layout & Facilities

The Table 1 - Inter distance between various facilities required at CGS has been added covering typical arrangement.

Cl.
5.2

Piping and Valves

- i) Minimum thickness of the piping has been specified.
- ii) Anti buoyancy measures have been specified for laying steel mains in areas normally under water or subject to flooding.

Cl.
5.2.2

Polyethylene (PE) Piping

Design formula has been mentioned for calculating the thickness with pressure rating of the pipeline.

Design pressure has been increased from **4 bar to 7 bar** as per existing regulations.

Network Isolation Valves

Cl.
5.2.4

Network Isolation Valves

- i) Valves of **size 4 inches and above** shall be fitted with double block and bleed facility.
- ii) Any valve shall be so positioned as to ensure unrestricted access for its operation and maintenance. Consideration shall be given to providing sufficient support for any valve.
- iii) All underground valves shall be provided with corrosion protection using **suitable anti-corrosion coating and sacrificial anodes** based on site conditions if required.

Consumer Installations and Proximity to Properties

Cl.
5.3

Consumer Installations:

Gas based water heater /Geyser and appliance valve shall be installed outside bathroom in fully ventilated space like terrace / balcony / passageways / Open veranda with adequately protected from the weather to avoid accident due to Carbon monoxide (CO) poisoning.

Cl.
5.6

Clause no 5.6: Proximity to Properties

No steel pipeline or main should be located within three (3) meters of any habitable dwelling or any industrial building unless it is provided with at least 300 mm of cover over and above minimum cover specified under Table-4 or special protective measures such as concrete slab, steel casing are provided.

Clause no 5.7 PNG Odorization

Cl.
5.7

Odour level tests shall be carried out to recognise the odour imparted by the gas supplier/distribution company. These tests are to be carried out at various defined locations on the network and at network extreme ends.

If odour level falls below the minimum acceptable level, same shall have to be intimated to the control room of the gas supplier and accordingly corrective actions are to be taken.

Fire Protection Facilities

Cl.
5-8

- i) Requirement of water sprinkler facility in gas handling area has been removed as per existing regulations.
- ii) Fire extinguisher requirement has been modified based on coverage area.

Depth of Cover and GIS

Cl.
6.2

Depth of Cover

Low depth covers at multi storey with basement area has been covered with risk mitigation measures.

Cl.
7.3

GIS implementation (New Clause)

The CGD operator shall put in place a GIS (Geographical Information System) with the objective of capturing the entire underground gas network and customer database.

Leakage survey & Integrity Management

Cl.
8.4

Clause 8.4: Leakage surveys (New Clause)

Operating company must have an effective method to identify and locate leakages in the system.

Highly congested areas shall be surveyed using gas detectors at least once in three months.

Other less congested areas shall be surveyed at least once a year.

Cl.
9.0

Clause 9.0: Integrity Management (New Clause)

Integrity Management program shall be documented and followed in line with existing regulations.



Table 1 - Inter distance between various facilities required at CGS

Sr. No.	From / To	1	2	3	4	5	6
1	Compound Wall	-	6	6	6	6	6
2	Control Room / Office Building / Store	6	-	12	12	2	15
3	Pressure Regulation and / or Metering	6	12	-	2	12	15
4	Odorant System	6	12	2	-	12	15
5	Electrical Sub Station	-#	2	12	12	-	15
6	Gas fired heaters	6	15	15	15	15	-

River course monitoring methodology

